

Rikin Bahadur Shrestha

Full-Stack Software Engineer

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Open to Relocation | Authorized to Work in the U.S.

Professional Summary

Full-Stack Software Engineer with hands-on experience building production tools in React, Node.js, FastAPI, and PostgreSQL at Projxon, and a research background in applied ML, including a co-authored paper accepted at IEEE ICECET 2026. BS in Computer Information Technology, Minnesota State University, Mankato, Dec. 2026.

Education

Minnesota State University, Mankato, BS in Computer Information Technology Jan. 2022 – Dec. 2026

- **Major:** Computer Information Technology **Certificate:** Graphic Design
- **Relevant Coursework:** Web/App Development, Networking, Cloud Systems, Databases, Machine Learning

Technical Skills

- **Languages:** Python, JavaScript, TypeScript, C#, C++, R
- **Frontend & Backend:** React.js, Tailwind CSS, CSS, Node.js, Express.js, FastAPI, REST APIs
- **Databases & Cloud:** PostgreSQL, MongoDB, SQL, AWS, Git, CI/CD Pipelines
- **Machine Learning & Data:** Scikit-learn, XGBoost, LightGBM, CatBoost, NLP, TF-IDF, Pandas, NumPy

Experience

APPLICATION DEVELOPER INTERN | Projxon May 2026 – Present

- Core developer on two internal platforms, **OrkaMIP**, intern milestone tracking, and **OrkaSupport**, a company-wide support ticketing platform, built end-to-end with **React.js**, **Node.js**, **Tailwind CSS**, **FastAPI**, and **PostgreSQL**.
- Took primary ownership of OrkaMIP's development within a small team, building its calendar-based progress tracking and milestone views for interns and managers; usage scales with the company's active intern headcount.
- Replaced a manual, spreadsheet-based ticketing process by building OrkaSupport per the company's requirements, introducing role-based dashboards and real-time status tracking now used across all **7 departments and 40+ teams**.
- Integrated REST APIs across both platforms to ensure consistent, role-based experiences for every member.

RESEARCH ASSISTANT | AIMS Research, MSU Jun. 2023 – Apr. 2026

- Built and evaluated **NLP** text classification pipelines across two Kaggle datasets (~160K and ~17K samples) using supervised learning models, assessed with **F1** and **ROC-AUC** metrics.
- Benchmarked three **gradient boosting models** (**XGBoost**, **LightGBM**, **CatBoost**) on a **TF-IDF** pipeline, achieving up to **96% accuracy and F1 0.74** on the larger dataset; first published results on these datasets.
- Co-authored a **published paper on ML-based cyberbullying detection**, accepted for publication at **IEEE ICECET 2026**, and presented findings at the National Conference on Undergraduate Research (**NCUR**).

Projects

2D SPACE SHOOTER | Unity | C#, OOP with Design Patterns Apr. 2026

- Architected the game around **11 software design patterns**, including **State** (player lifecycle), **Strategy** (enemy movement), **Factory** and **Prototype** (enemy creation), **Facade** (level orchestration), **Command** (input actions), and **Observer** (UI reactivity), for a fully decoupled, extensible architecture.
- Built a custom **object-pooling system** for player and enemy projectiles to eliminate per-shot instantiation overhead and sustain stable frame rates during heavy-fire combat.

INTERNSHIP DATABASE MANAGEMENT SYSTEM | SQL, Normalization & Optimization Mar. 2026

- Designed a **relational database schema** applying **normalization** to reduce redundancy, improve structural integrity.
- Optimized **SQL queries and indexing** strategies to improve query performance and reduce server load.

SERVICE REQUEST MANAGEMENT SYSTEM (SRMS) | MERN Stack Dec. 2025

- Planned the application workflow and led **frontend development** of a MERN-based service request platform, designing the **CRUD** flows and access model for creating, viewing, updating, and managing requests.
- Built a responsive **React interface** with **role-based access control**, tailoring views by user type.

Certifications & Publications

AWS Certified Cloud Practitioner, In Progress

Full-Stack Web Development with MERN Stack

Aug. 2025

Cyberbullying Detection Across Unequal Datasets: A Gradient Boosting Model Comparison, IEEE ICECET 2026